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DETECTION SYSTEM, DETECTION METHOD, AND RECORDING MEDIUM

Abstract

A detection system includes: an internal communication monitor that at least monitors a first communication performed from a terminal to a device; a determiner that determines whether the first communication monitored by the internal communication monitor includes an attack made on the device by the terminal; and an outputter that outputs information indicating a result of the determining by the determiner. When the determiner determines that the first communication is from the terminal to an ephemeral port of the device, the determiner determines that the first communication includes the attack.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This is a continuation application of PCT International Application No. PCT/JP2023/038562 filed on Oct. 25, 2023, designating the United States of America, which is based on and claims priority of Japanese Patent Application No. 2022-193259 filed on Dec. 2, 2022. The entire disclosures of the above-identified applications, including the specifications, drawings and claims are incorporated herein by reference in their entirety.

FIELD

[0002] The present disclosure relates to a detection system, a detection method, and a recording medium.

BACKGROUND

[0003] Internet of things (IoT), such as household electrical appliances, have been used. Furthermore, measures against security attacks made on IoT devices have been disclosed (see Patent Literature (PTL) 1, for example).

CITATION LIST

Patent Literature

[0004] PTL 1: Japanese Unexamined Patent Application Publication No. 2017-175462

SUMMARY

Technical Problem

[0005] Unfortunately, a security attack made on an IoT device via communication can go undetected by a method of detecting a security attack made on, for example, a conventional server device.

[0006] In response to this, the present disclosure provides a detection system and so forth that appropriately detect a security attack made on a device via communication.

Solution to Problem

[0007] A detection system according to an aspect of the present disclosure includes: a monitor that at least monitors a first communication performed from a terminal to a client device; a determiner that determines whether the first communication monitored by the monitor includes an attack made on the client device by the terminal; and an outputter that outputs information indicating a result of the determining by the determiner, wherein when the determiner determines that the first communication is from the terminal to an ephemeral port of the client device, the determiner determines that the first communication includes the attack.

[0008] The general or specific aspect described above may be realized by a device, a method, an integrated circuit, a computer program, or a computer-readable recording medium such as a Compact Disc Read-Only Memory (CD-ROM) or may be realized by any combination of the device, the method, the integrated circuit, the computer program, and the recording medium.

Advantageous Effects

[0009] The detection system according to the present disclosure is capable of appropriately detecting a security attack made on a device via communication.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0010] These and other advantages and features will become apparent from the following description thereof taken in conjunction with the accompanying Drawings, by way of non-limiting examples of embodiments disclosed herein.

[0011] FIG. 1 is an explanatory schematic diagram of a configuration of a detection system and a communication system that includes the detection system, according to Embodiment.

[0012] FIG. 2 is an explanatory diagram illustrating an app list used by the detection system, according to Embodiment.

[0013] FIG. 3 is a flowchart of processing relating to communication monitoring performed by the detection system, according to Embodiment.

[0014] FIG. 4 is an explanatory diagram illustrating an authorized-server list according to Embodiment.

[0015] FIG. 5 is an explanatory diagram illustrating an unauthorized-server list according to Embodiment.

[0016] FIG. 6 is a flowchart of processing performed by the detection system to detect a host scan, according to Embodiment.

[0017] FIG. 7 is a flowchart of processing performed by the detection system to determine an ephemeral port, according to Embodiment.

[0018] FIG. 8 is an explanatory diagram illustrating ephemeral ports of a device according to Embodiment.

[0019] FIG. 9 is a flowchart of processing performed by an internal communication monitor to detect an attack and wiretapping, according to Embodiment.

DESCRIPTION OF EMBODIMENT

Underlying Knowledge Forming Basis of the Present Disclosure

[0020] In relation to the technology for security measures in the Background section, the inventors have found the following issue.

[0021] A conventional security attack on, for example, a server device is made by transmission of a communication packet (also referred to simply as a packet) to a transmission control protocol (TCP) port or a user datagram protocol (UDP) port (also referred to simply as a port) used by the server device for service provision. Note that a security attack is also commonly called a cyberattack.

[0022] The port used by the server device for service provision is basically managed as a well-known port by the Internet Assigned Numbers Authority (IANA), and is predetermined. For example, TCP port **80** is used by a hypertext transfer protocol (HTTP) server (or, a so-called web server) for service provision (that is, for web-page provision). Furthermore, TCP ports **20** and **21** are used by a file transfer protocol (FTP) server for service provision (that is, for file transfer and for control over the file transfer).

[0023] On this account, to detect a security attack made on a conventional server, it is useful to verify whether a communication device performs a communication that includes an attack against a well-known port of the server. To detect such a security attack, a system called a honeypot has been used. The honeypot is a system that detects a security attack by connecting a possible attack-target server device to the Internet and, while exposing this server device to an attacker from the Internet, monitoring communication performed on the aforementioned well-known port.

[0024] In recent years, IoT devices, such as household electrical appliances connected to networks,

have been used. An IoT device is capable of operating by communicating with a terminal (a smartphone, for example) that controls the IoT device or with a server.

[0025] A security attack can possibly be made on the IoT device via communication. The security attack on the IoT device via the communication is made by application software (also referred to simply as an app) unauthorizedly installed on the terminal that controls the IoT device, or made by an unauthorized server, for example. Such a security attack causes the IoT device to perform an unauthorized operation. In response to this, the measures against security attacks on IoT devices have been disclosed (see PTL 1, for example).

[0026] However, the method of detecting a security attack on a conventional server may be unable to detect the security attack on the IoT device. The IoT device is connected to the server via a port (also commonly referred to as an ephemeral port or a dynamic port) that is temporarily used for connection to the server, instead of via a well-known port. For this reason, even when whether the communication device performs a communication that includes an attack on the well-known port of the IoT device is verified, the presence or absence of a security attack on the IoT device cannot be determined. Note that the port used as the ephemeral port is selected from among ports managed as ephemeral ports by the IANA.

[0027] In view of this, the present disclosure provides a detection system and so forth that appropriately detect a security attack made on a device via communication.

[0028] The following describes an example of the disclosure obtained from the disclosure of the present specification and also describes advantageous effects obtained from the disclosure.

[0029] (1) A detection system includes: a monitor that at least monitors a first communication performed from a terminal to a client device; a determiner that determines whether the first communication monitored by the monitor includes an attack made on the client device by the terminal; and an outputter that outputs information indicating a result of the determining by the determiner, wherein when the determiner determines that the first communication is from the terminal to an ephemeral port of the client device, the determiner determines that the first communication includes the attack.

[0030] According to the aspect described above, when the communication is performed from the terminal to the ephemeral port of the client device, the detection system determines that the terminal has made a security attack (also referred to simply as an attack) on the client device. Thus, the detection system can detect the attack that is undetected even when the communication performed on a well-known port of the client device is monitored. In this way, the detection system is capable of appropriately detecting the security attack made on the device via the communication.

[0031] (2) The detection system according to (1), wherein the monitor further monitors a second communication that is from the client device to a communication device that is connected via an external network, and when the second communication includes a communication from the client device to a predetermined authorized server, the determiner further obtains, from the client device, a port of the client device that is used in the second communication, and determines, based on the port as the ephemeral port, whether the first communication includes the attack.

[0032] According to the aspect described above, the detection system is capable of detecting the attack made by the terminal on the ephemeral port that is actually used by the client device to communicate with the authorized server. In this way, the detection system is capable of appropriately detecting the security attack made on the ephemeral port that is actually used by the device.

[0033] (3) The detection system according to claim (2), wherein the monitor further monitors a third communication between the terminal and a communication device that is connected via an external network, and the determiner further determines whether the third communication includes a communication performed from the terminal to an unauthorized server that is different from the predetermined authorized server, and determines that the third communication relates to the attack when the determiner determines that the third communication includes the communication

performed from the terminal to the unauthorized server.

[0034] According to the aspect described above, based on the communication performed on the unauthorized server by the terminal, it is detected that the terminal has performed the communication relating to the attack. This terminal that is detected to have performed the communication relating to the attack has a relatively high probability of making an attack on the client device. Thus, the detection system is capable of appropriately detecting the security attack made on the device via the communication.

[0035] (4) The detection system according to any one of (1) to (3), wherein the monitor further monitors an internal communication performed from the terminal to a destination having an address included in an internal network, and the determiner further determines whether the internal communication includes a packet relating to a host scan for the internal network.

[0036] According to the aspect described above, the detection system is capable of detecting that the terminal has performed the host scan. This terminal that is detected to have performed the host scan has a relatively high probability of making an attack on the client device. This is because the communication device found by the host scan can possibly be targeted for an attack. Thus, as described above, the detection system is capable of more appropriately detecting the security attack made on the device via the communication.

[0037] (5) The detection system according to (4), wherein when the determiner determines that the first communication includes the attack and that the internal communication does not include the packet relating to the host scan, the determiner determines that the terminal is suspected of a wiretapping attack.

[0038] According to the aspect described above, it is possible to detect that the terminal has made the attack on the client device without performing the host scan. This terminal is suspected of the security attack executed through communication wiretapping. Thus, the detection system is capable of more appropriately detecting the security attack made on the device via the communication.

[0039] (6) The detection system according to (2), wherein the monitor further monitors an internal communication performed from the terminal to a destination having an address included in an internal network, and the determiner further determines whether the internal communication includes a packet relating to a host scan for the internal network, and when the determiner determines that the first communication includes the attack, that the internal communication does not include the packet relating to the host scan, and that a time difference between the communication from the client device to the predetermined authorized server and the attack included in the first communication is within a reference time period, determines that the terminal is suspected of a wiretapping attack.

[0040] According to the aspect described above, it is possible to detect that the terminal has made the attack on the client device without performing the host scan and that the terminal has made the attack on the client device within a relatively short time after the communication with the authorized server. This terminal is suspected of the security attack executed through communication wiretapping. Thus, the detection system is capable of more appropriately detecting the security attack made on the device via the communication.

[0041] (7) The detection system according to any one of (1) to (6), wherein application software for performing the first communication is installed on the terminal, and when the determiner determines that the first communication includes the attack, the determiner determines that the application software has a function of making an attack.

[0042] According to the aspect described above, the detection system is capable of detecting the app that has made the attack on the client device. In this way, the detection system is capable of appropriately detecting the app that makes the security attack on the device via the communication.

[0043] (8) A detection method executed by a detection system to detect an attack, the detection method including: at least monitoring a first communication performed from a terminal to a client device; determining whether the first communication monitored in the monitoring includes an

attack made on the client device by the terminal; and outputting information indicating a result of the determining, wherein in the determining, when it is determined that the first communication is from the terminal to an ephemeral port of the client device, it is determined that the first communication includes the attack.

[0044] The above aspect produces the same advantageous effects as those produced by the above-described detection system.

[0045] (9) A non-transitory computer-readable recording medium having recorded thereon a program for causing a computer to execute the detection method according to (8).

[0046] The above aspect produces the same advantageous effects as those produced by the above-described detection system.

[0047] Each of the general or specific aspects described above may be realized by a device, a method, an integrated circuit, a computer program, or a computer-readable recording medium such as a CD-ROM or may be realized by any combination of the device, the method, the integrated circuit, the computer program, and the recording medium.

[0048] Hereinafter, certain exemplary embodiments will be described in detail with reference to the accompanying Drawings.

[0049] The following embodiments are general or specific examples of the present disclosure. The numerical values, shapes, materials, constituent elements, arrangement and connection configuration of the constituent elements, steps, the order of the steps, etc., described in the following embodiment are merely examples, and are not intended to limit the present disclosure. Among constituent elements in the following embodiment, those not described in any one of the independent claims indicating the broadest concept of the present disclosure are described as optional constituent elements.

EMBODIMENT

[0050] The following describes a detection system and so forth that appropriately detect a security attack made on a device via communication according to the present embodiment.

[0051] FIG. 1 is an explanatory schematic diagram of a configuration of detection system **10** and communication system **1** that includes detection system **10** according to the present embodiment.

[0052] Communication system **1** detects a security attack that can be possibly made on device **21** by terminal **22**. Communication system **1** is first described.

[0053] As illustrated in FIG. 1, communication system **1** includes base station **20**, device **21**, and terminal **22**. Communication system **1** is connected to authorized server **31** and unauthorized server **32** via network **N2**.

[0054] Base station **20** is a communication device that enables device **21** and terminal **22** to be communicatively connected to network **N2**. For example, base station **20** is an access point of a wireless local area network (wireless LAN) (such as Wi-Fi (registered trademark)) installed in a residence. This case is described as an example.

[0055] Base station **20** is connected to network **N2** via a communication line (or more specifically, a wired LAN), and is wirelessly connected to device **21** and terminal **22**. Base station **20** has a function of transferring packets sent and received between devices connected. Device **21** or terminal **22** occasionally communicates with authorized server **31** or unauthorized server **32** via network **N2**. Note that whether base station **20** is connected to device **21**, terminal **22**, and network **N2** by wire or wirelessly is not limited to the example described above. To be more specific, base station **20** may be connected to device **21** or terminal **22** by wire (that is, by a communication line), and base station **20** may be connected to network **N2** wirelessly.

[0056] Device **21** fulfills a function unique to device **21** while communicating with terminal **22** and authorized server **31** via base station **20**. Device **21** corresponds to a client device. For example, device **21** is a household electrical appliance, such as an air purifier or a washing machine. Note that device **21** may be not only a household electrical appliance but also a client device connected to a typical server via communication.

[0057] Terminal **22** is an information terminal that provides device **21** with information via base station **20** to cause device **21** to operate. For example, terminal **22** is a smartphone, a tablet, or a personal computer. Terminal **22** includes application software (also referred to simply as app) **23** that runs on an operation system (OS). App **23** can be authorized app **231** or unauthorized app **232**. Note that although the drawing shows that both authorized app **231** and unauthorized app **232** operate on terminal **22** for the purpose of description, at least one of authorized app **231** or unauthorized app **232** may operate on terminal **22**.

[0058] Authorized app **231** collects information from device **21** or provides device **21** with an operation instruction or information. Note that authorized app **231** does not necessarily need to communicate directly with device **21**. Authorized app **231** may collect the information from device **21** or provide device **21** with the operation instruction or the information, via authorized server **31**. In this case, authorized app **231** does not directly communicate with device **21** and thus communicates only with authorized server **31**.

[0059] In contrast, unauthorized app **232** makes a security attack on device **21**, for example. Examples of the security attack made by unauthorized app **232** can include: unauthorizedly collecting information from device **21** by disguising itself as authorized app **231** or authorized server **31** to communicate with device **21**; and causing device **21** to malfunction by providing device **21** with an instruction or information for an unauthorized operation. Thus, detection system **10** may be a system that detects a security attack that can be made on device **21** by unauthorized app **232**. In other words, unauthorized app **232** is targeted for determination made by detection system **10** as to whether unauthorized app **232** makes a security attack. It is possible to say that unauthorized app **232** is targeted for security-attack detection by detection system **10**.

[0060] Note that the number of terminals **22** may be two or more.

[0061] Note that any number of unauthorized apps **232** may operate on terminal **22**. However, it is useful to set the number of unauthorized apps **232** operating on terminal **22** at one because, when detection system **10** detects a security attack, this one unauthorized app **232** is identified as the app that makes the security attack. The present embodiment describes the example where the number of unauthorized apps **232** operating on terminal **22** is one.

[0062] Note that when the number of unauthorized apps **232** operating on terminal **22** is at least two and detection system **10** detects a security attack, at least one of the at least two unauthorized apps **232** operating on terminal **22** is identified as unauthorized app **232** that makes a security attack. For the case where a plurality of unauthorized apps **232** operating on one terminal **22** make a security attack collaboratively, detection system **10** is able to detect the security attack by keeping the plurality of unauthorized apps **232** operating on one terminal **22**. For terminal **22** installed with single unauthorized app **232**, this method is effective against unauthorized app **232** that avoids detection by detection system **10** by keeping itself hidden without making a security attack. Furthermore, this method is also effective when the presence of authorized app **231** triggers unauthorized app **232** to make a security attack.

[0063] Authorized server **31** controls operation of device **21** or provides device **21** with information. Authorized server **31** controls the operation of device **21** by providing device **21** with an instruction for normal operation of device **21** via communication. Furthermore, authorized server **31** obtains information on normal operation of device **21** with authorization, from device **21** via communication. Furthermore, authorized server **31** receives an operation instruction to device **21** for running authorized app **231**, by communicating with authorized app **231**. Furthermore, by providing authorized app **231** with the information on the operation obtained from device **21** with authorization, authorized server **31** is able to present the operation information on authorized app **231**. Authorized server **31** is assumed to be prepared by a business operator that provides device **21** and authorized app **231**.

[0064] Furthermore, authorized server **31** provides, via communication, information (for example, update information on firmware or an app of the terminal, or time information) used in operations

of device **21** and authorized app **231**. Authorized server **31** may be prepared by the business operator that provides device **21** and app **23**, or may be prepared to provide information to a wide range of common terminals (such as a time server that provides time information to a wide range of common terminals according to the network time protocol (NTP)). Thus, a plurality of authorized servers **31** may be included corresponding to roles provided.

[0065] Unauthorized server **32** is different from authorized server **31**. For example, unauthorized server **32** relates to operation of unauthorized app **232**. This case is described as an example. In this case, unauthorized server **32** provides an instruction for unauthorized operation of unauthorized app **232** via communication with terminal **22**. Furthermore, unauthorized server **32** obtains information on unauthorized operation of device **21** that is unauthorizedly obtained by unauthorized app **232**, from terminal **22** via communication. Unauthorized server **32** corresponds to a server (a so-called command-and-control (C&C) server) that controls terminal **22** on which unauthorized app **232** operates. Unauthorized server **32** is assumed to be prepared by someone that is different from the business operator providing device **21** and authorized app **231** and that intentionally causes device **21** to perform an unauthorized operation.

[0066] Note that examples of unauthorized server **32** include a server different from a server registered as authorized server **31**. On this account, the examples of unauthorized server **32** may include not only a server relating to an unauthorized operation of device **21** but also a server that is not registered as authorized server **31** (such as an app that is not intended to be widely used by ordinary people or an app that is not widely used by ordinary people).

[0067] Detection system **10** monitors communication on a communication line that connects base station **20** to network N2 and communication on network N1. Network N2 is a network outside the residence, and examples include the Internet. Network N1 is a network inside the residence. By monitoring the aforementioned communications, detection system **10** is able to detect a security attack that can be made by terminal **22** (or app **23**), or more specifically, detect that app **23** is unauthorized app **232**.

[0068] Detection system **10** is next described.

[0069] As illustrated in FIG. 1, detection system **10** includes functional sections as external communication monitor **11**, internal communication monitor **12**, determiner **13**, storage **14**, and outputter **15**. Each of the functions of external communication monitor **11**, internal communication monitor **12**, determiner **13**, and outputter **15** can be implemented in part or in whole by a processor (a central processing unit (CPU)) included in detection system **10** and executing a predetermined program using a memory.

[0070] External communication monitor **11** monitors external communication (also referred to as a second communication). The external communication refers to communication on the communication line that connects base station **20** to network N2. More specifically, the external communication refers to communication between device **21** or terminal **22** and authorized server **31** or unauthorized server **32**. External communication monitor **11** is disposed on the communication line that connects base station **20** to network N2. External communication monitor **11** captures a packet transferred over the communication line and provides determiner **13** with captured data on the packet.

[0071] External communication monitor **11** corresponds to one function of a monitor. To be more specific, external communication monitor **11** corresponds to a monitor that monitors the communication (also referred to as the second communication) from device **21** to the communication device connected via network N2.

[0072] Internal communication monitor **12** monitors internal communication. The internal communication refers to communication performed wirelessly by base station **20** on network N1. More specifically, the internal communication refers to communication between base station **20** and device **21** or terminal **22**. In other words, the internal communication refers to communication on network N1. Internal communication monitor **12** is disposed in a position where radio waves

transmitted to or from base station **20** is receivable (or more specifically, a position in a region in which base station **20** is enabled to perform wireless communication). Internal communication monitor **12** captures a packet transmitted or received by device **21** or terminal **22** on network N1 and provides determiner **13** with captured data on the packet.

[0073] Internal communication monitor **12** corresponds to one function of the monitor. To be more specific, internal communication monitor **12** corresponds to a monitor that at least monitors communication (also referred to as a first communication) from terminal **22** to device **21**.

[0074] Determiner **13** determines whether the communication monitored by external communication monitor **11** or internal communication monitor **12** includes a security attack.

Determiner **13** obtains, from external communication monitor **11**, the captured data on the packet transferred on the communication line that connects base station **20** to network N2. Moreover, determiner **13** obtains, from internal communication monitor **12**, the captured data on the packet transmitted or received over network N1. Then, determiner **13** makes the aforementioned determination based on the captured data obtained.

[0075] To be more specific, when determining, based on the captured data on the packet transmitted or received over network N1, that the first communication is performed from terminal **22** to an ephemeral port of device **21**, determiner **13** determines that the first communication includes the security attack. At this time, determiner **13** can also determine that app **23** installed on terminal **22** has a function of making a security attack.

[0076] An ephemeral port is commonly a TCP port or a UDP port of the communication device and is temporarily used when the communication device is connected to a server device according to the TCP or the UDP. The ephemeral port is selected from among ports with numbers within a predetermined range. This range is from **49152** to **65535** for example, and is managed by the IANA.

[0077] Furthermore, when determining whether the first communication includes an attack, determiner **13** can use the captured data on the packet transferred on the communication line that connects base station **20** to network N2. More specifically, when the communication from device **21** to authorized server **31** is included in the second communication, determiner **13** obtains the port of device **21** that is used in the second communication. Then, determiner **13** determines whether the first communication includes an attack, based on the port obtained as the ephemeral port.

[0078] Determiner **13** corresponds to a determiner that determines whether the first communication monitored by the monitor includes an attack made on device **21** by terminal **22**.

[0079] Storage **14** is a volatile or nonvolatile storage device capable of storing information used in the processing performed by determiner **13** and information indicating a result of the determination made by determiner **13**. For example, storage **14** is a random access memory (RAM), a hard disk drive (HDD), or a solid state drive (SSD).

[0080] Storage **14** stores an app list (see FIG. 2) as the information used in the processing performed by determiner **13** and the information indicating a result of the determination made by determiner **13**. Furthermore, storage **14** stores a port list (see FIG. 8) indicating the ephemeral ports of device **21**.

[0081] Outputter **15** outputs result information indicating the result of the determination stored in storage **14**. The result information outputted by outputter **15** includes at least information indicating the presence or absence of the attack made by terminal **22** (see FIG. 2). Furthermore, the result information outputted by outputter **15** may include: the presence or absence of communication with the unauthorized server; the presence or absence of scan execution; the presence or absence of suspicion of wiretapping execution; or a time difference (see FIG. 2).

[0082] Outputter **15** includes a presentation device, for example, and presents the result information to the user via the presentation device. The presentation of the result information can include: displaying the result information on a display screen that is an example of the presentation device; or outputting the result information by voice from a loudspeaker that is an example of the

presentation device. Note that outputter **15** may include a communication interface. In this case, outputter **15** may provide the result information to an external presentation device via the communication interface and then cause the external presentation device to present the result information.

[0083] As described above, the determination by determiner **13** that the first communication includes the security attack corresponds to the security attack detection by detection system **10**. Furthermore, the determination by determiner **13** that app **23** has the function of making a security attack corresponds to the detection by detection system **10** that app **23** is unauthorized app **232**.

[0084] Note that the functional sections of detection system **10** may be included in a single device or separately located in a plurality of devices. For the functional sections of detection system **10** that are separately located in the plurality of devices, the information is transmitted or received via the communication interface between devices that need the information.

[0085] FIG. **2** is an explanatory diagram illustrating the app list used by detection system **10**, according to Embodiment. The app list illustrated in FIG. **2** indicates at least one app that is targeted for security attack detection by detection system **10**. In other words, this app list indicates an app that is targeted for detection as to whether the app makes a security attack. The app list is stored in storage **14**. The content of the app list can be updated by determiner **13**.

[0086] The app list illustrated in FIG. **2** shows, for each of the at least one app: internet protocol (IP) address; name; the presence or absence of communication with unauthorized server; the presence or absence of scan execution; the presence or absence of attack execution; the presence or absence of suspicion of wiretapping execution; and time difference.

[0087] The IP address indicates an IP address of terminal **22** that is installed with the present app. Note that an identifier that uniquely identifies terminal **22** can be used instead of the IP address.

[0088] The name indicates a name of the present app. The name refers to attached textual information on the app.

[0089] The presence or absence of communication with the unauthorized server is information indicating whether the present app has communicated with unauthorized server **32**. When determiner **13** determines that terminal **22** installed with the present app has communicated with unauthorized server **32**, the information on the presence or absence of communication with the unauthorized server is updated to information indicating that the present app has communicated with unauthorized server **32** (that is, the information indicating the presence of communication with unauthorized server **32**). Although the information indicating the presence of communication with unauthorized server **32** is indicated by a checkmark in FIG. **2**, the form of the information is not limited to this. Likewise, the forms of information described later are not intended to be limiting.

[0090] Note that the presence or absence of communication with the unauthorized server merely indicates the result of the determination by determiner **13**. Thus, from this information, the server with which the app has communicated cannot be determined to be an unauthorized server.

[0091] The presence or absence of scan execution is information indicating whether the present app has executed a host scan. When determiner **13** determines that terminal **22** installed with the present app has executed the host scan, the information on the presence or absence of scan execution is updated to information indicating that the present app has executed the host scan (that is, information indicating the presence of scan execution).

[0092] The presence or absence of attack execution is information indicating whether the present app has executed a security attack. When determiner **13** determines that terminal **22** installed with the present app has executed the security attack, the information on the presence or absence of attack execution is updated to information indicating that the present app has executed the security attack (that is, the information indicating the presence of attack execution).

[0093] The presence or absence of suspicion of wiretapping execution is information indicating whether the present app is suspected of wiretapping the communication. When determiner **13** determines that terminal **22** installed with the present app is suspected of wiretapping the

communication, the information on the presence or absence of suspicion of wiretapping execution is updated to information indicating that the present app is suspected of wiretapping the communication (that is, the information indicating the presence of suspicion of wiretapping execution).

[0094] The time difference indicates, when the present app is suspected of wiretapping the communication, a difference between a time of authorized communication and a time of attack execution.

[0095] Determiner **13** updates the information of the app list according to the result of the determination made based on the captured data. The content of the app list at the time when determiner **13** finishes the determination indicates the result of the determination by detection system **10**.

[0096] To be more specific, the app list illustrated in FIG. **2** includes four apps with numbers #1 to #4.

[0097] App #1 has the name “CLUB-P” and is installed on terminal **22** having the IP address “192.168.1.2”. Determiner **13** determines that app #1 communicates with an unauthorized server. App #1 is an example of an app used in information exchange in a group of people or an organization. App #1 is not intended to be widely used by ordinary people and is an example of an unknown server that is not registered as authorized server **31**.

[0098] App #2 has the name “Wi-Fi Analyze” and is installed on terminal **22** having the IP address “192.168.1.3”. Determiner **13** determines that app #1 communicates with an unauthorized server and that app #1 has executed a host scan. App #2 is a tool for analyzing Wi-Fi communication. App #2 is not intended to be widely used by ordinary people and is intended to be used by a person with knowledge and skills relating to communication.

[0099] App #3 has the name “Gooogle Map” and is installed on terminal **22** having the IP address “192.168.1.4”. Determiner **13** determines that app #3 communicates with an unauthorized server and that app #3 has executed a host scan and also has executed a security attack.

[0100] App #4 has the name “Pachimon Go” and is installed on terminal **22** having the IP address “192.168.1.5”. Determiner **13** determines that app #4 communicates with an unauthorized server and that app #4 has executed a security attack and is suspected of wiretapping execution.

[0101] Hereafter, the processing performed by detection system **10** is described.

[0102] FIG. **3** is a flowchart of processing relating to communication monitoring performed by detection system **10** according to the present embodiment. FIG. **4** is an explanatory diagram illustrating an authorized-server list according to the present embodiment. FIG. **5** is an explanatory diagram illustrating an unauthorized-server list according to the present embodiment. The processing relating to communication monitoring performed by external communication monitor **11** is described with reference to FIG. **3**, FIG. **4**, and FIG. **5**.

[0103] In Step **S101**, determiner **13** determines whether communication between terminal **22** and an external device (this communication corresponds to the external communication) is detected. To be more specific, determiner **13** obtains, from external communication monitor **11**, capture data on packets transferred over the communication line that connects base station **20** to network **N2**. When determining that the packets include a packet relating to the communication between terminal **22** and the external device, determiner **13** determines that the external communication is detected. The external device can be authorized server **31** or unauthorized server **32**. When it is determined that the external communication is detected (Yes in Step **S101**), the processing proceeds to Step **S102**. Otherwise (No in Step **S101**), Step **S101** is executed again.

[0104] In Step **S102**, determiner **13** determines whether a communication partner of the external communication targeted for the determination in Step **S101** is unauthorized server **32**. This determination can be made by using the authorized-server list (see FIG. **4**). The authorized-server list includes an IP address (such as 123.45.67.89 in the example illustrated in FIG. **4**) of at least one server that is previously approved as authorized server **31**. When the IP address of the

communication partner of the external communication targeted for the determination in Step S101 is not included in the authorized-server list, determiner 13 determines that the communication partner of the external communication is unauthorized server 32. When it is determined that the communication partner of the external communication is unauthorized server 32 (Yes in Step S102), the processing proceeds to Step S103. Otherwise (No in Step S102), Step S101 is executed again.

[0105] Note that authorized server 31 can be the communication partner of the communication performed by terminal 22 that is in an initial state. The initial state refers to a state where an OS and an app provided together with the OS have been installed and app 23 has yet to be installed. Examples of the server that can be previously approved as authorized server 31 can include: a server that distributes an app that runs on the OS; and a server that distributes time information. By determining the communication partner of the communication performed by terminal 22 in the initial state, determiner 13 is able to generate the authorized-server list in advance. Note that, together with or in place of the above, the authorized-server list may include a server approved as an authorized server by an administrator.

[0106] In Step S103, determiner 13 determines that the external communication targeted for the determination in Step S101 is the communication between terminal 22 and unauthorized server 32. Moreover, determiner 13 determines that the communication partner of the external communication is unauthorized server 32. Furthermore, determiner 13 determines that this communication relates to a security attack and thus registers the communication partner of this external communication into the unauthorized-server list (see FIG. 5). The unauthorized-server list includes an IP address (such as 98.76.54.32 in the example illustrated in FIG. 5) of a server determined to be an unauthorized server. The communication between terminal 22 and unauthorized server 32 can include: issuance of an unauthorized-operation instruction from unauthorized server 32 to terminal 22; or provision of unauthorized-operation information from terminal 22 to unauthorized server 32. The unauthorized-operation information includes the IP address or model name of device 21 found in a port scan (described later) performed by terminal 22, for example.

[0107] In Step S104, determiner 13 determines that app 23 installed on terminal 22 is an unauthorized-server communication app. Determiner 13 updates the information on the presence or absence of communication with the unauthorized server in the app list to information indicating that app 23 has communicated with unauthorized server 32.

[0108] After the end of Step S104, Step S101 is executed again.

[0109] FIG. 6 is a flowchart of processing performed by detection system 10 to detect a host scan, according to the present embodiment.

[0110] In Step S201, determiner 13 determines whether communication in which a transmission source is terminal 22 is detected. To be more specific, determiner 13 obtains, from internal communication monitor 12, capture data on a packet transmitted or received over network N1. When determining that the transmission source of the packet is terminal 22, determiner 13 determines that the communication in which the transmission source is terminal 22 is detected.

[0111] In Step S202, determiner 13 determines whether the communication detected in Step S201 is performed for a host scan. When it is determined that this communication is performed for the host scan (Yes in Step S202), the processing proceeds to Step S203. Otherwise (No in Step S202), Step S201 is executed again.

[0112] A host scan can be performed for the purpose of finding a communication device present within network N1. More specifically, a host scan can be performed by an unauthorized app for the purpose of finding an attack target.

[0113] One example of the host scan is a ping scan. A ping scan is performed by sending an echo request (a so-called ping request) using an internet control message protocol (ICMP) echo request message sequentially to at least one destination corresponding to at least one IP address that can be within network N1. Note that another example of the host scan is performed by using an address

resolution protocol (ARP) request.

[0114] In Step S203, determiner 13 determines that app 23 installed on terminal 22 is a scan execution app. Moreover, determiner 13 determines a time of the executed host scan. Determiner 13 updates the information on the presence or absence of a scan execution app to information indicating that app 23 has executed the scan.

[0115] After the end of Step S203, Step S201 is executed again.

[0116] FIG. 7 is a flowchart of processing performed by detection system 10 to determine an ephemeral port, according to the present embodiment. FIG. 8 is an explanatory diagram illustrating a port list indicating the ephemeral ports of device 21 according to the present embodiment. The processing performed by detection system 10 to determine the ephemeral port is described with reference to FIG. 7 and FIG. 8.

[0117] In Step S301, determiner 13 determines whether communication between device 21 and authorized server 31 (this communication corresponds to the authorized communication) is detected. To be more specific, determiner 13 obtains, from external communication monitor 11, capture data on packets transferred over the communication line that connects base station 20 to network N2. When determining that the packets include a packet relating to the communication between device 21 and authorized server 31, determiner 13 determines that the communication between device 21 and authorized server 31 is detected. The communication between device 21 and authorized server 31 can include: issuance of an authorized-operation instruction from authorized server 31 to device 21; or provision of authorized-operation information from device 21 to authorized server 31. When it is determined that the communication between device 21 and authorized server 31 is detected (Yes in Step S301), the processing proceeds to Step S302. Otherwise (No in Step S301), Step S301 is executed again.

[0118] In Step S302, determiner 13 determines, as the ephemeral port, the port used by device 21 in the communication detected in Step S301. Moreover, determiner 13 determines a time of the communication detected in Step S301. Determiner 13 registers the determined ephemeral port and the time of the communication into the port list (see FIG. 8). Note that when the identical ephemeral port is already registered in the port list, the time corresponding to this ephemeral port is updated.

[0119] The port list illustrated in FIG. 8 indicates that device 21 has communicated with authorized server 31 using the ephemeral port with number 49513 at 07:05:03.001 on Jan. 5, 2022, for example.

[0120] Note that the port list is updated whenever the authorized communication is detected in Step S301. Of the information included in the port list, information obtained a predetermined time period or longer before the current time may be deleted by determiner 13. Although the predetermined time period may be substantially from several seconds to several minutes for example, this is not intended to be limiting.

[0121] Note that when a plurality of devices 21 are present, the port list is prepared for each of the plurality of devices 21.

[0122] FIG. 9 is a flowchart of processing performed by internal communication monitor 12 to detect an attack and wiretapping, according to the present embodiment.

[0123] In Step S401, determiner 13 determines whether communication from terminal 22 to device 21 is detected. To be more specific, determiner 13 obtains, from internal communication monitor 12, capture data on a packet transmitted or received over network N1. When determining that a transmission source of the packet is terminal 22 and that the destination of the packet is device 21, determiner 13 determines that the communication from terminal 22 to device 21 is detected. When it is determined that the communication from terminal 22 to device 21 is detected (Yes in Step S401), the processing proceeds to Step S402. Otherwise (No in Step S401), Step S401 is executed again.

[0124] In Step S402, determiner 13 determines whether the destination port of the communication

targeted for the determination in Step S401 is an ephemeral port of device 21. This determination can be made by using the port list indicating the ephemeral ports of device 21 (see FIG. 8). Determiner 13 determines whether the destination port of the communication targeted for the determination in Step S401 is included among the ephemeral ports of device 21 in the communication log. When it is determined that the destination port of the communication is the ephemeral port of the device 21 (Yes in Step S402), the processing proceeds to Step S403. Otherwise (No in Step S402), Step S401 is executed again.

[0125] In Step S403, determiner 13 determines whether the time of the communication performed using the ephemeral port targeted for the determination in Step S402 as the destination port is within a reference time period after the time of the latest authorized communication. The time of the latest authorized communication may be obtained from the time of the communication performed using the ephemeral port that is included in the port list of device 21 illustrated in FIG. 8 and that is targeted for the determination in Step S402. Although the reference time period may be about one second for example, this is not intended to be limiting. When it is determined that the time of the communication is within the reference time period after the time of the latest authorized communication (Yes in Step S403), the processing proceeds to Step S404. Otherwise (No in Step S403), Step S401 is executed again.

[0126] Note that Step S403 need not be performed. When Step S403 is not executed and determiner 13 determines that the destination port of the communication is the ephemeral port of device 21 in Step S402 (Yes in Step S402), the processing proceeds to Step S404.

[0127] In Step S404, determiner 13 determines that app 23 installed on terminal 22 is a security-attack execution app, or more specifically, an app having a function of making a security attack. In this case, the communication performed using the ephemeral port targeted for the determination in Step S402 as the destination port corresponds to a security attack. Determiner 13 updates the information on the presence or absence of an attack execution app in the app list to information indicating that app 23 has made the security attack.

[0128] Note that the aforementioned security attack is assumed to be triggered when terminal 22 (or app 23) finds the latest authorized communication and then determines the ephemeral port of device 21 used in this authorized communication. To define the security attack as being triggered by the authorized communication depending on how much time has passed after this authorized communication, the reference time period used in Step S403 can be adjusted.

[0129] In Step S405, determiner 13 determines whether the time of the security attack is within a reference time period after the last host scan performed before this security attack. The time of the security attack is the time of the communication in which the ephemeral port targeted for the determination in Step S402 is used as the destination port. The time of the host scan is the time at which the communication to perform the host scan is detected (Step S202 and Step S203 in FIG. 6). Although the reference time period may be substantially from several minutes to several hours for example, this is not intended to be limiting. When it is determined that the time of the security attack is within the reference time period after the last host scan (Yes in Step S405), Step S401 is executed again. Otherwise (No in Step S405), the processing proceeds to Step S406.

[0130] In Step S406, determiner 13 determines that app 23 installed on terminal 22 is suspected of wiretapping execution, or more specifically, that app 23 is an app suspected of having a wiretapping function. This is because, as it is determined in Step S404 and Step S405 that terminal 22 (or app 23) has made the security attack without performing the host scan, terminal 22 (or app 23) is highly likely to find, by wiretapping, device 21 that is an attack target. Determiner 13 updates the information on the presence or absence of an app suspected of wiretapping execution in the app list to information indicating that app 23 is suspected of wiretapping the communication.

[0131] After the end of Step S406, Step S401 is executed again.

[0132] The processes of Step S404 and Step S405 allow determiner 13 to determine that the first communication includes an attack and, when it is determined that the internal communication does

not include the packet relating to the host scan, to determine that terminal **22** is suspected of the wiretapping attack.

[0133] Furthermore, the processes of Step **S403** to Step **S405** allow determiner **13** to determine that the first communication includes the attack and that the internal communication does not include the packet relating to the host scan. Furthermore, the processes allow determiner **13** to determine, when it is determined that the time difference between the communication from device **21** to authorized server **31** and the attack included in the first communication is within the reference time period, that terminal **22** is suspected of the wiretapping attack.

[0134] As described thus far, detection system **10** is capable of appropriately detecting the security attack made on device **21** by terminal **22** (or app **23**) via the communication.

[0135] Each of the constituent elements in the above embodiment may be configured in the form of an exclusive hardware product, or may be realized by executing a software program suitable for the constituent element. Each of the constituent elements may be realized by means of a program executing unit, such as a Central Processing Unit (CPU) or a processor, reading and executing the software program recorded on a recording medium such as a hard disk or semiconductor memory.

[0136] To be more specific, this program causes a computer to execute a detection method that is executed by a detection system to detect an attack. The detection method includes: at least monitoring a first communication performed from a terminal to a client device; making a determination whether the first communication monitored in the monitoring includes an attack made on the client device by the terminal; and outputting information indicating a result of the determination. In the making, when it is determined that the first communication is from the terminal to an ephemeral port of the client device, it is determined that the first communication includes the attack.

[0137] Although the detection system and the like according to one or more aspects of the present disclosure have been described based on the embodiment, the present disclosure is not limited to this embodiment. Those skilled in the art will readily appreciate that embodiments arrived at by making various modifications to the above embodiment or embodiments arrived at by selectively combining constituent elements disclosed in the above embodiment without materially departing from the scope of the present disclosure may be included within one or more aspects of the present disclosure.

INDUSTRIAL APPLICABILITY

[0138] The present disclosure is applicable to a system that detects an app that executes a security attack.

Claims

1. A detection system comprising: a monitor that at least monitors a first communication performed from a terminal to a client device; a determiner that determines whether the first communication monitored by the monitor includes an attack made on the client device by the terminal; and an outputter that outputs information indicating a result of the determining by the determiner, wherein when the determiner determines that the first communication is from the terminal to an ephemeral port of the client device, the determiner determines that the first communication includes the attack.
2. The detection system according to claim 1, wherein the monitor further monitors a second communication that is from the client device to a communication device that is connected via an external network, and when the second communication includes a communication from the client device to a predetermined authorized server, the determiner further obtains, from the client device, a port of the client device that is used in the second communication, and determines, based on the port as the ephemeral port, whether the first communication includes the attack.
3. The detection system according to claim 2, wherein the monitor further monitors a third communication between the terminal and a communication device that is connected via an external

network, and the determiner further determines whether the third communication includes a communication performed from the terminal to an unauthorized server that is different from the predetermined authorized server, and determines that the third communication relates to the attack when the determiner determines that the third communication includes the communication performed from the terminal to the unauthorized server.

4. The detection system according to claim 1, wherein the monitor further monitors an internal communication performed from the terminal to a destination having an address included in an internal network, and the determiner further determines whether the internal communication includes a packet relating to a host scan for the internal network.

5. The detection system according to claim 4, wherein when the determiner determines that the first communication includes the attack and that the internal communication does not include the packet relating to the host scan, the determiner determines that the terminal is suspected of a wiretapping attack.

6. The detection system according to claim 2, wherein the monitor further monitors an internal communication performed from the terminal to a destination having an address included in an internal network, and the determiner further determines whether the internal communication includes a packet relating to a host scan for the internal network, and when the determiner determines that the first communication includes the attack, that the internal communication does not include the packet relating to the host scan, and that a time difference between the communication from the client device to the predetermined authorized server and the attack included in the first communication is within a reference time period, determines that the terminal is suspected of a wiretapping attack.

7. The detection system according to claim 1, wherein application software for performing the first communication is installed on the terminal, and when the determiner determines that the first communication includes the attack, the determiner determines that the application software has a function of making an attack.

8. A detection method executed by a detection system to detect an attack, the detection method comprising: at least monitoring a first communication performed from a terminal to a client device; determining whether the first communication monitored in the monitoring includes an attack made on the client device by the terminal; and outputting information indicating a result of the determining, wherein in the determining, when it is determined that the first communication is from the terminal to an ephemeral port of the client device, it is determined that the first communication includes the attack.

9. A non-transitory computer-readable recording medium having recorded thereon a program for causing a computer to execute the detection method according to claim 8.
